

# Virtual Feed: A Simulated Breastfeeding Experience in Virtual Reality

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Figure 1: First person view of an virtual avatar holding, feeding and caressing a virtual baby

## ABSTRACT

Technology use in antenatal education is often basic and inadequately presents the emotional and practical challenges of breastfeeding. This work presents *Virtual Feed*, a Virtual Reality interactive breastfeeding simulation designed with parents and health care professionals to convey key aspects of breastfeeding and to explore the potential and limitation of interactive technology in eliciting intimacy in parent-child space.

## CCS CONCEPTS

• **Software and its engineering** → *Software prototyping*.

## KEYWORDS

breastfeeding, virtual reality

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## 1 INTRODUCTION AND BACKGROUND

Breastfeeding is beneficial for parents and children, but the process can be challenging, with many parents feeling insufficiently prepared for the reality of breastfeeding. Current antenatal education predominantly focuses on presenting the benefits of breastfeeding and detailing the feeding process, e.g., by employing info-graphics, printed materials, passive media and probes. Technology use in breastfeeding education is often basic [14], and fails to display the lived experience in terms of practical and emotional challenges adequately [15]. Prior work [4, 5, 14] has speculated about the potential of interactive systems and Virtual Reality (VR) to improve breastfeeding education. Here, we present *Virtual Feed*, a VR based interactive breastfeeding simulation that portrays breastfeeding on the basis of parents' lived experience of infant feeding, the input of expert stakeholders, and breastfeeding literature. Our work aims to give expecting parents a more realistic glimpse of key elements of breastfeeding, and further explores the potential and limitations of VR to portray breastfeeding as an intimate and emotional process.

### 1.1 Immersive Interactive Systems and Breastfeeding Education

Little research has explored the potential of VR in the context of parenting and antenatal education. Hsieh et al. [5] explored the potential of VR in conveying parental love through a VR experience that leverages a tangible probe representing a baby and an algorithm that generates a baby face similar to the user; the authors suggest future research to explore the potential of VR in supporting

nursing education and simulating the experience of bottle feeding. Yanardag [18] developed a VR breastfeeding experience which transports parents into a soothing environment to support parents when expressing milk; the status of the work is, however, unknown. Generally, we see that one of the biggest challenges for designing interactive breastfeeding experiences is to find an adequate representation of the bond between the virtual child and the parent, creating a simulation that engages users by fostering relatedness [17].

## 1.2 Conveying Parent-Child Intimacy

In the HCI community, intimacy is often vaguely defined [13] and not specifically tailored to the relationship between parent and child. Scholars associate intimacy with the act of self revealing [11, 12], sharing of the self and being vulnerable to one another [7], reciprocal feelings of trust and emotional closeness toward each other [16]. Drawing from these, we frame intimacy as the closeness and connectedness between parents and their child through the act of nurturing. We structure our work around this definition and set to convey intimacy through delicate interactions such as holding and caressing and portrayal of newborns' vulnerable behaviours. Interestingly, intimacy is one of the breastfeeding facilitators that nursing literature often neglects [3], despite parents highlighting its importance for the establishment of a successful feeding relationship.

## 1.3 Presenting Human Characters

Presenting human characters or humanoid artefacts in (interactive) media is challenging. Our perception may experience discomfort when confronted with almost realistic humanoid objects, as explained by the uncanny valley theory [9]. In interactive-immersive systems, the eeriness effect goes beyond the aesthetic, touching upon the (embodied) interaction [2], motion and animation [6]. A common strategy to steer away from this pitfall is to present an abstract representation and avoid embracing the limitations of technology. In the context of breastfeeding simulations, this suggests that great care needs to be taken to create digital representations of a baby that can foster relatedness and engage parents in a positive way.

## 2 DESIGN AND DEVELOPMENT OF VIRTUAL FEED

Virtual Feed is a VR breastfeeding experience in which the user is invited to breastfeed a newborn baby in various places by means of holding a physical representation of a baby doll. The system described here is an experience prototype [8] that serves as conversation starter to engage relevant stakeholders in conversations about VR to support breastfeeding, and will be refined on the basis of focus group that further explore visual styles and scenario design. It was designed in consultation with care4education, an antenatal education provider in Flanders.

### 2.1 Hardware and Software Platforms

The simulation consists of three hardware components (see figure 2): 1) Oculus Rift CV1 head mounted display (HMD) and its constellations tracking sensors, 2) a doll representing a newborn attached

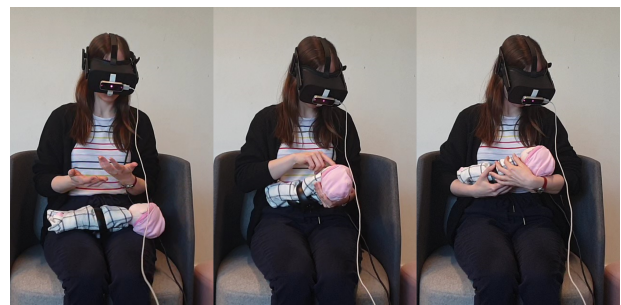


**Figure 2: Physical representation of the baby, VR head mounted device and sensors for hands and motion tracking**

with an Oculus Touch controller (for pose estimation and motion tracking), and 3) a Leap Motion hand tracking controller to enable hand interactions. In terms of software development platform, Unity is used together with software development kits provided by Oculus and Leap motion.

### 2.2 Embodied Avatar and Interactions

We choose to present a full body avatar to enhance users' immersion and sense of embodiment [10], i.e., improving the sense of self-location, body-ownership and agency. From a technical perspective, motion tracking capability of the hardware enables parts of the player's upper body to be tracked; positions of forearm, arm, neck and spine that are not tracked, are estimated using Forward And Backward Reaching Inverse Kinematic (FABRIK) [1]. This allows us to portray an embodied avatar in a sitting pose (see figure 1) with support for full upper-body movement. The simulation incorporates three essential interactions in breastfeeding:



**Figure 3: Interactions within the simulation (From left to right): free hand interaction, caressing and holding**

**Caressing the Baby** Free hand interaction adds degree of freedoms that improve task performance and embodiment [10], and facilitates more delicate interactions with the baby, such as touching and caressing.

**Holding and Feeding** interactions further improve the closeness, inviting the player to hold the probe close to their body. This interaction, however, often disrupts the free hand interaction due to the limitations of the hand tracking device. We overcome this by estimating the hand pose using FABRIK [1].

**Latching and Unlatching** processes portray the initiation and completion of feeding. Here, users are required to interpret the baby's behaviours and initiate the latch process by bringing the baby's mouth in proximity of the breast. We also animate facial expressions of the virtual baby.

### 2.3 Look and Feel

In terms of aesthetic within the VR space, we apply a simple low-polygon visual style when crafting 3D models (see figure 1, 5 and 6). This style reduces the heavy computation/rendering workload and can potentially steer the experience away from the uncanny valley. This way, a plausible user experience is maintained.

In terms of the physical representation of the baby, we opt for a soft off-the-shelf plush doll from Haba that resembles the dimensions of a newborn while providing an abstract presentation of the baby. We insert an extra weight (of ~1.5kg) inside the doll to give the player a realistic impression of actual newborn weight. The final version of the doll will incorporate additional options for skin tones and hair colour to serve a diverse community, i.e., we will provide additional dolls that are equipped with the motion tracking sensors. These options can also be adjusted to match the presentation within the simulation.

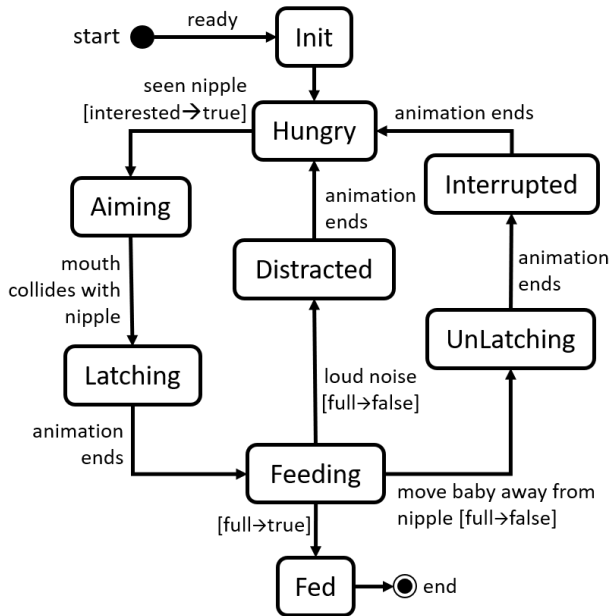


Figure 4: state-machine-diagram modelling the behaviours of the virtual baby

### 2.4 Feeding mechanic

The feeding mechanic is designed to convey a typical experience of breastfeeding. Here, we crafted a morphing 3D model, animations and audio cues that communicate the virtual newborns' response to player interactions. The feeding mechanic and the behaviors of the baby is presented in a state-machine-diagram, composing of 9 states (see figure 4). Each state can be triggered by a player input event (e.g., aiming the baby's mouth at breast), a time out event of the previous state, or pre-programmed events in the virtual space (e.g., an impulsive noise to distract the baby).

### 2.5 Breastfeeding Scenarios

Two breastfeeding scenarios were crafted to convey the feeding experience in distinct environments, each echoing nuances of influences from environmental factors with the goal of prompting users to reflect on breastfeeding.

**Scenario one:** a personal living space was crafted to demonstrate a pleasant feeding experience in a familiar environment. We depict a breastfeeding scene in a living room (see figure 5) that has basic furniture, appliances, newborn related objects and background noises from home appliances.



Figure 5: Scenario 1 - Personal living space

**Scenario two:** a public park aims to manifest the perceived environmental and social influences when breastfeeding in public spaces (see figure 6). In this scenario, the user breastfeeds on a bench in a public park, in which other people are carrying out leisure activities (e.g., jogging).



Figure 6: Scenario 2 - Public park

## 3 EXPECTED EXPERIENCE

We expect the conference attendees to experience the intimacy and the delicate nature of breastfeeding through our VR application. We

hope that the crafted scenarios can help the conference attendees to reflect on how social norms and environmental factors influence parents' feeding practice. Moreover, we expect our simulation to spark reflection on potential pitfalls in conveying intimacy in immersive systems, for example, the critical balancing act between technical limitations, aesthetic styles and level of realism. More generally, we hope that conference attendees will be engaged in discussions about the potential of interactive technology to support education in potentially sensitive settings, and about the appropriateness of different forms of interactive media that could be deployed in our context.

## 4 DISCUSSION AND CONCLUSION

This paper describes the development of Virtual Feed, a VR breastfeeding simulation. The developed artefact will serve as a tool to probe inputs from (expecting) parents and healthcare professionals, allowing us to design a simulation aligned with parents' lived experiences of breastfeeding that are suited to supplement breastfeeding education. We hope that the demonstration of our work invites the HCI community to reflect on not only the user experience but also on the challenges when designing immersive technology in such a delicate setting.

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